# Orbital Box Worksheet

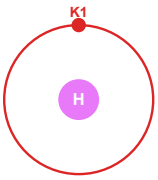
Name: _____ Class: _____ Section: _____ Date: November 07, 2026

Write ↑ or ↓ in each box, one arrow per electron. Use ↑↓ only when the orbital already holds one electron.

## Z = 1 H · Hydrogen

period 1 · group 1 · s-block · 1 valence electrons

**Bohr model**




1s

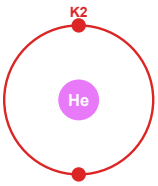



cut here

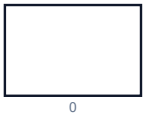
## Z = 2 He · Helium

period 1 · group 18 · s-block · 2 valence electrons

**Bohr model**




1s



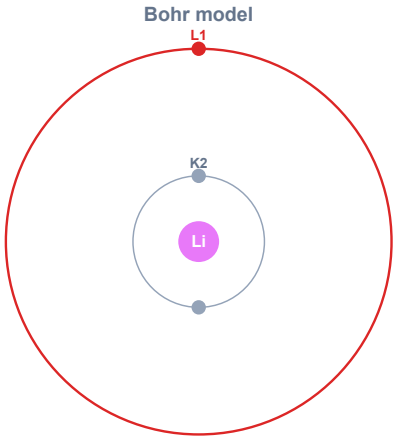
# Orbital Box Worksheet

Name: _____ Class: _____ Section: _____ Date: November 07, 2026

Write ↑ or ↓ in each box, one arrow per electron. Use ↑↓ only when the orbital already holds one electron.

## Z = 3 Li · Lithium

period 2 · group 1 · s-block · 1 valence electrons




1s

0

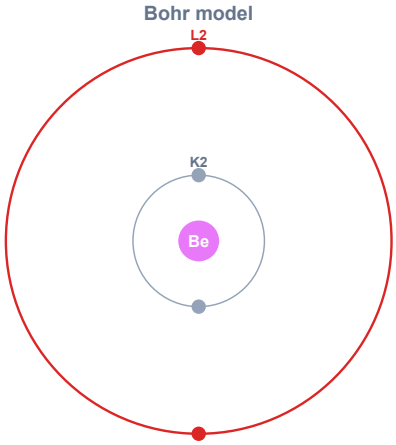
2s

0

cut here

## Z = 4 Be · Beryllium

period 2 · group 2 · s-block · 2 valence electrons




1s

0

2s

0

# Orbital Box Worksheet

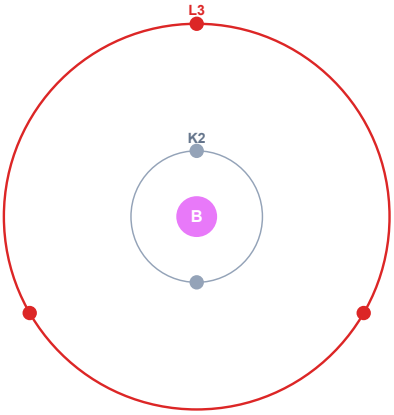
Name: _____ Class: _____ Section: _____ Date: November 07, 2026

Write ↑ or ↓ in each box, one arrow per electron. Use ↑↓ only when the orbital already holds one electron.

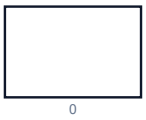
## Z = 5 B · Boron

period 2 · group 13 · p-block · 3 valence electrons

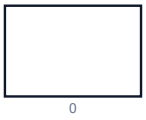
Bohr model



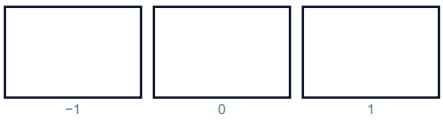

1s




2s




2p

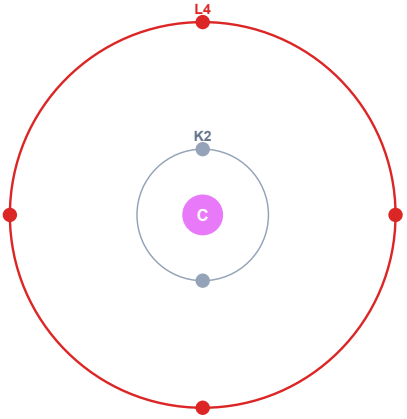



cut here

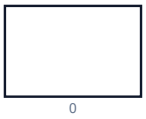
## Z = 6 C · Carbon

period 2 · group 14 · p-block · 4 valence electrons

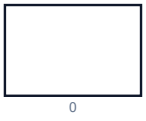
Bohr model



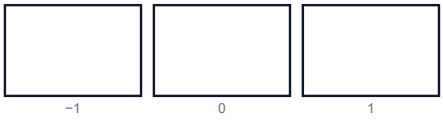

1s




2s




2p



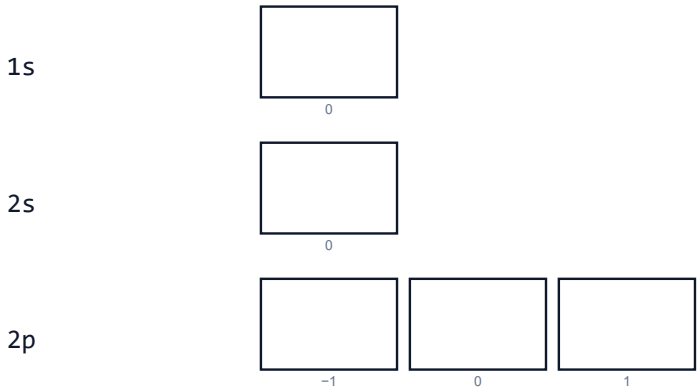
# Orbital Box Worksheet

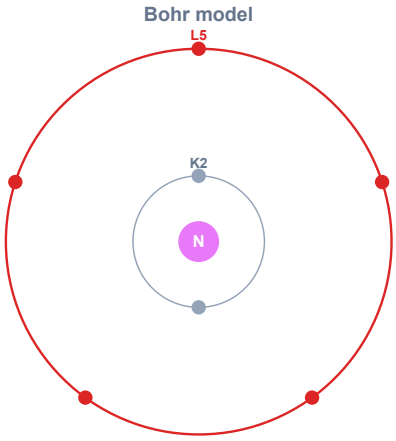
Name: _____ Class: _____ Section: _____ Date: November 07, 2026

Write ↑ or ↓ in each box, one arrow per electron. Use ↑↓ only when the orbital already holds one electron.

## Z = 7 N · Nitrogen

period 2 · group 15 · p-block · 5 valence electrons

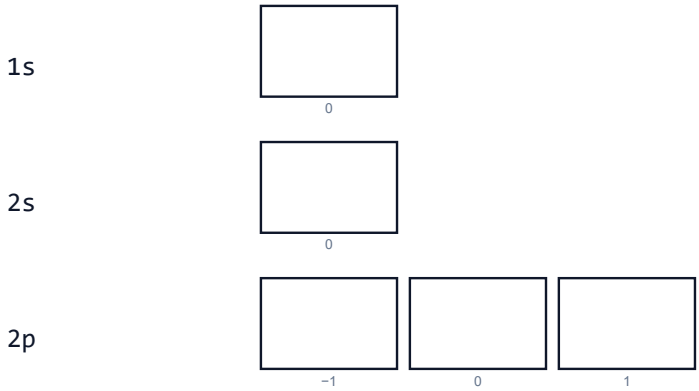


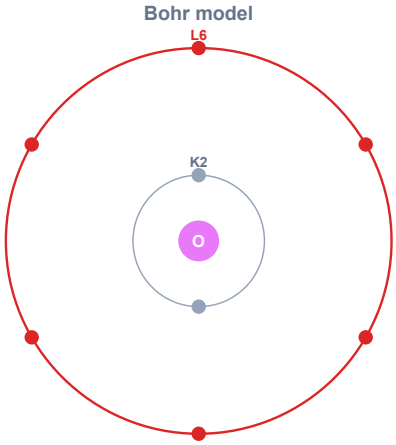




cut here

## Z = 8 O · Oxygen

period 2 · group 16 · p-block · 6 valence electrons



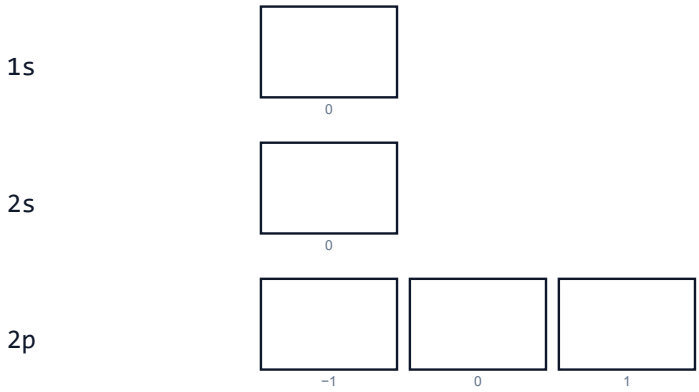
# Orbital Box Worksheet

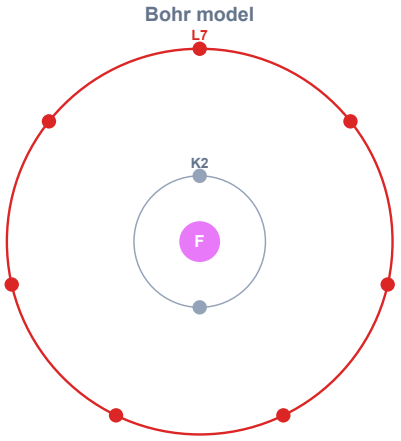
Name: _____ Class: _____ Section: _____ Date: November 07, 2026

Write ↑ or ↓ in each box, one arrow per electron. Use ↑↓ only when the orbital already holds one electron.

## Z = 9 F · Fluorine

period 2 · group 17 · p-block · 7 valence electrons

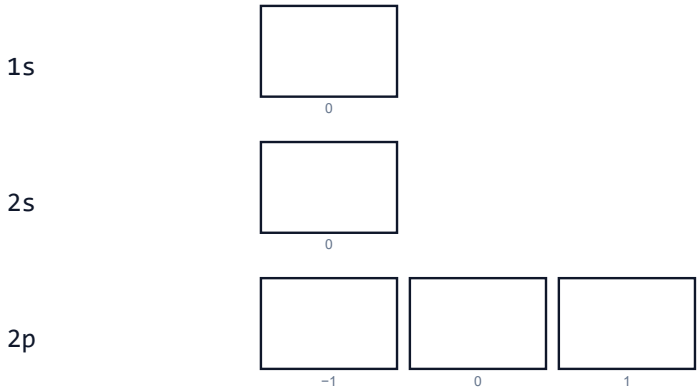


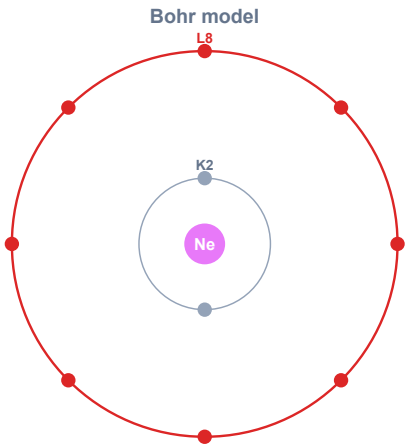




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## Z = 10 Ne · Neon

period 2 · group 18 · p-block · 8 valence electrons



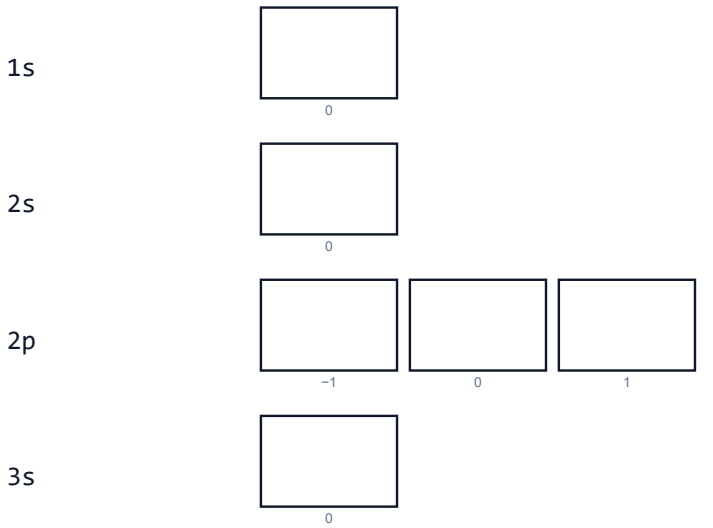
# Orbital Box Worksheet

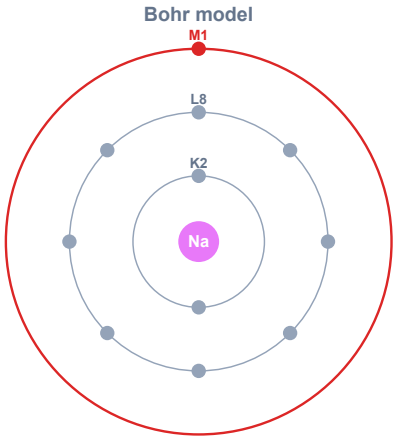
Name: _____ Class: _____ Section: _____ Date: November 07, 2026

Write ↑ or ↓ in each box, one arrow per electron. Use ↑↓ only when the orbital already holds one electron.

## Z = 11 Na · Sodium

period 3 · group 1 · s-block · 1 valence electrons

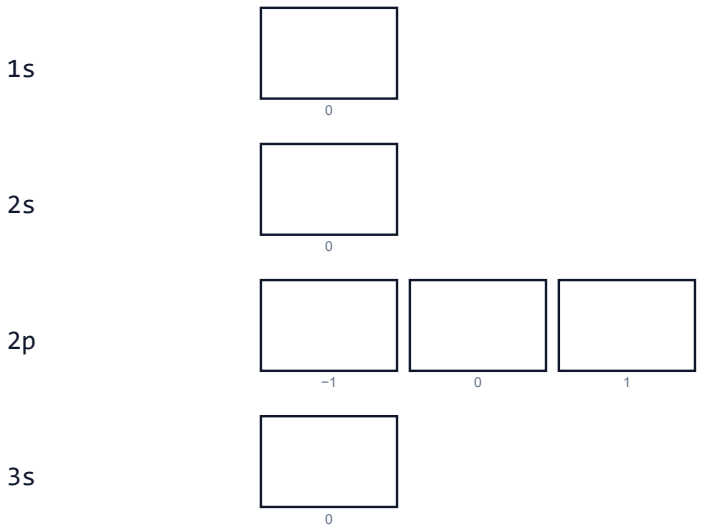


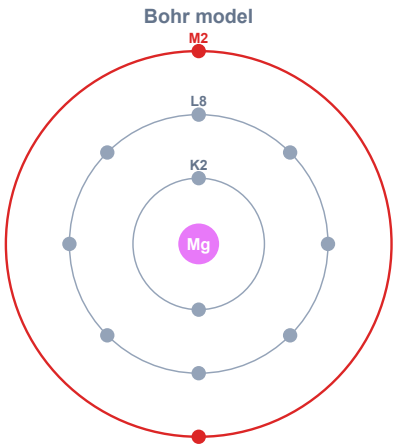




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## Z = 12 Mg · Magnesium

period 3 · group 2 · s-block · 2 valence electrons



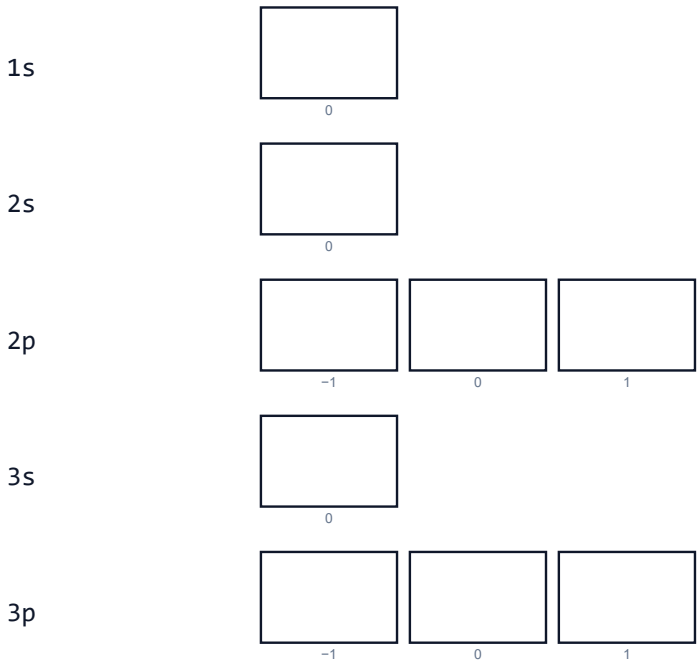
# Orbital Box Worksheet

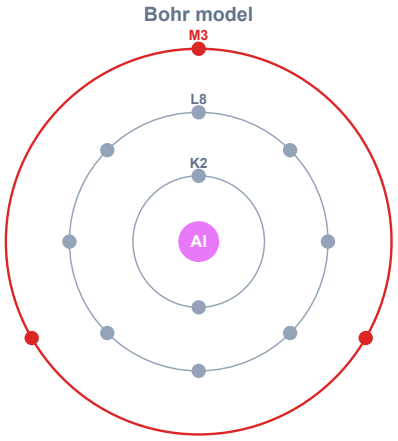
Name: _____ Class: _____ Section: _____ Date: November 07, 2026

Write ↑ or ↓ in each box, one arrow per electron. Use ↑↓ only when the orbital already holds one electron.

## Z = 13 Al · Aluminium

period 3 · group 13 · p-block · 3 valence electrons

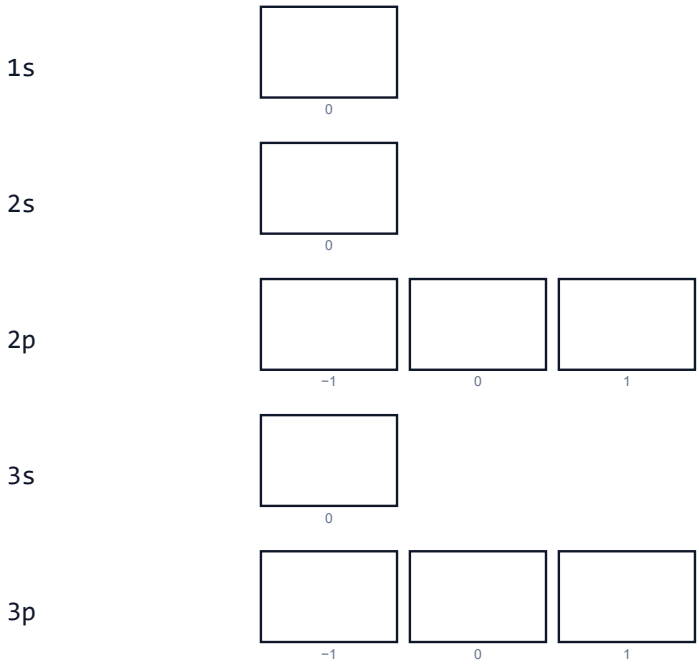


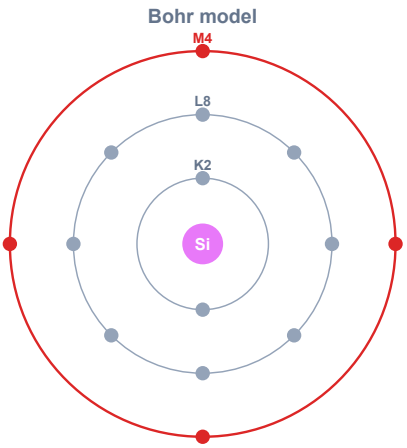




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## Z = 14 Si · Silicon

period 3 · group 14 · p-block · 4 valence electrons



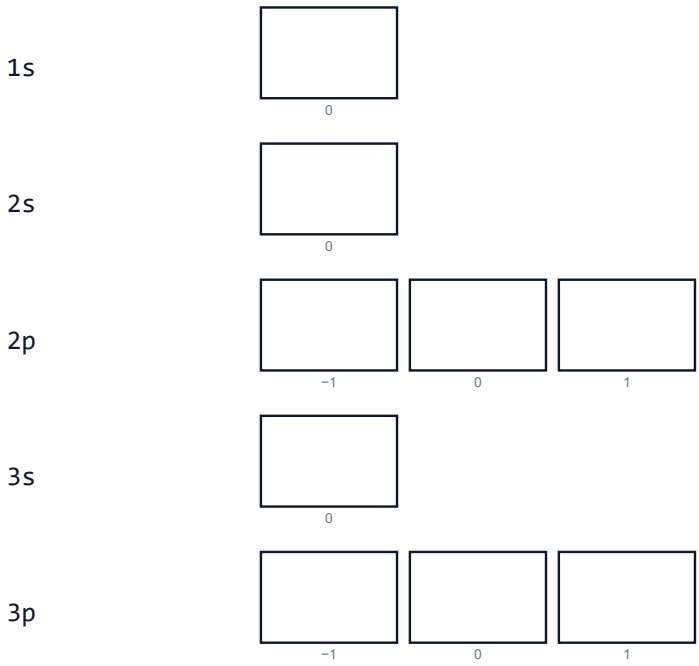
# Orbital Box Worksheet

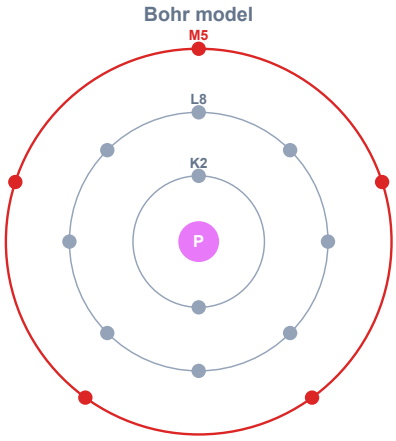
Name: _____ Class: _____ Section: _____ Date: November 07, 2026

Write ↑ or ↓ in each box, one arrow per electron. Use ↑↓ only when the orbital already holds one electron.

## Z = 15 P · Phosphorus

period 3 · group 15 · p-block · 5 valence electrons

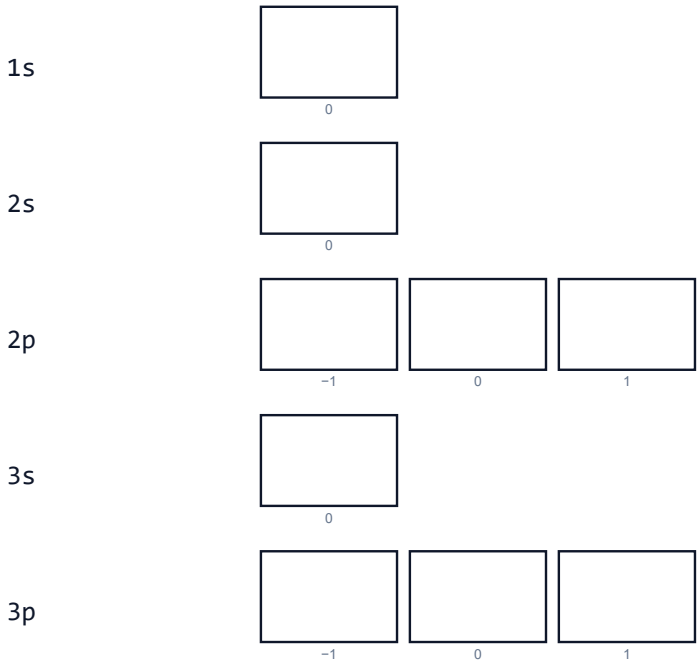


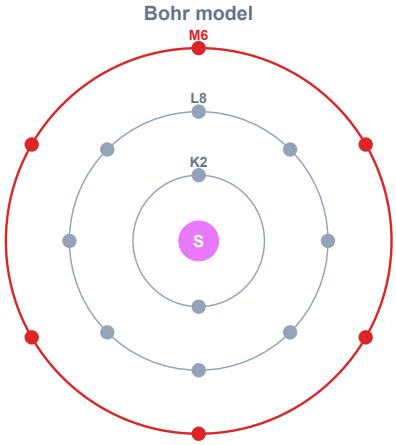




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## Z = 16 S · Sulfur

period 3 · group 16 · p-block · 6 valence electrons



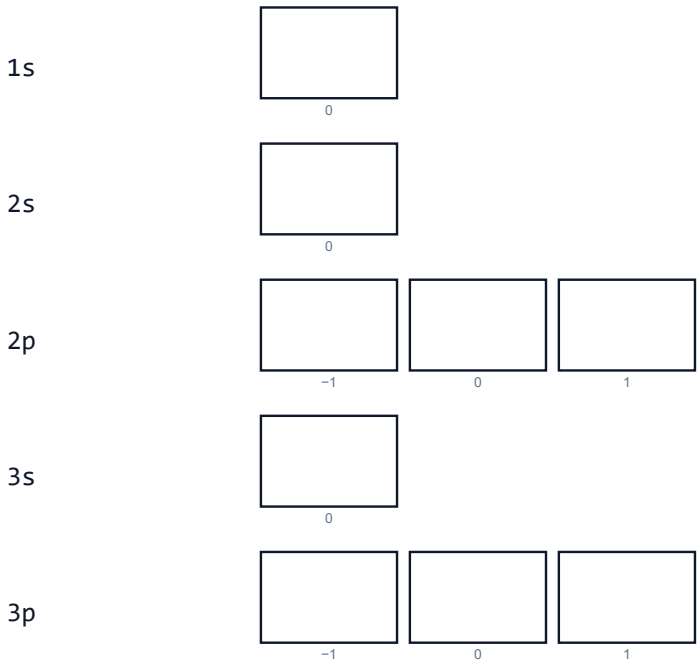
# Orbital Box Worksheet

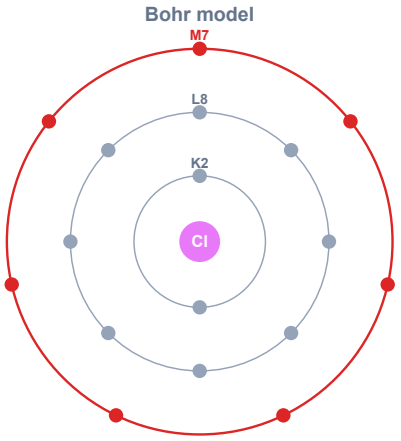
Name: _____ Class: _____ Section: _____ Date: November 07, 2026

Write ↑ or ↓ in each box, one arrow per electron. Use ↑↓ only when the orbital already holds one electron.

## Z = 17 Cl · Chlorine

period 3 · group 17 · p-block · 7 valence electrons

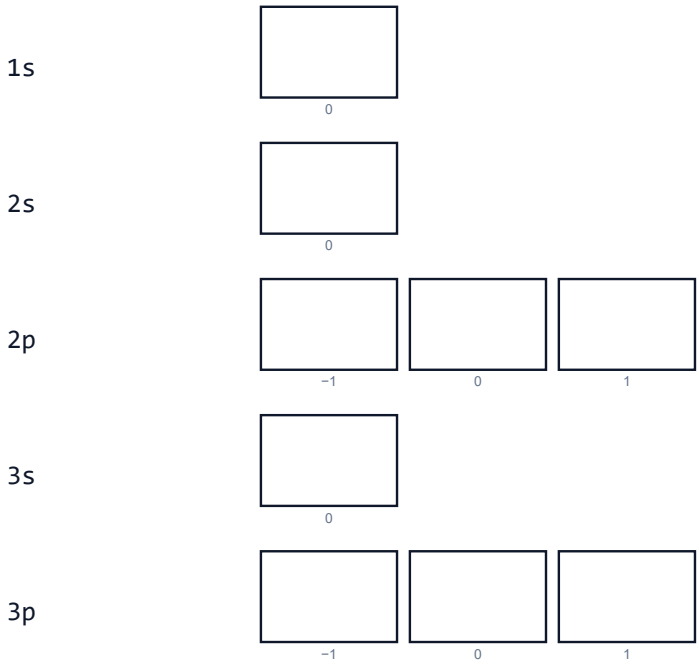


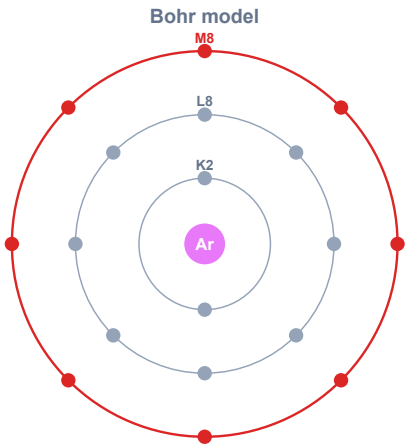




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## Z = 18 Ar · Argon

period 3 · group 18 · p-block · 8 valence electrons



# Orbital Box Worksheet

Name: _____ Class: _____ Section: _____ Date: November 07, 2026

Write ↑ or ↓ in each box, one arrow per electron. Use ↑↓ only when the orbital already holds one electron.

## Z = 19 K · Potassium

period 4 · group 1 · s-block · 1 valence electrons

1s
0

2s
0

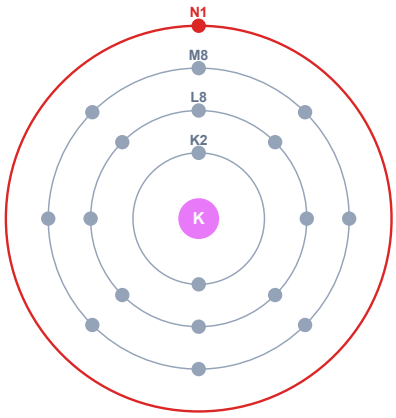
2p
−1 0 1

3s
0

3p
−1 0 1

4s
0

Bohr model




cut here

## Z = 20 Ca · Calcium

period 4 · group 2 · s-block · 2 valence electrons

1s
0

2s
0

2p
−1 0 1

3s
0

3p
−1 0 1

4s
0

Bohr model

